

Majorization-Minimization for Grouped L_0 Regularization in GLMs

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Motivation & Key Contributions

The Problem: Variable selection identifies covariate subsets for interpretable and predictive models.

Method	Pros	Cons
Lasso L_1 -norm	Convex Computationally tractable	Biased estimates Shrinks true signal
Best Subset Selection L_0 -norm	Unbiased Oracle property (Shen et al., 2013)	NP-hard Non-convex and discontinuous

Our Approach: To solve the complexity of best subset selection, we adopt the Majorization-Minimization (MM) approximation of L_0 introduced by Yen (2011).

Key Contributions:

- Tractable Optimization:** Just as EM algorithms bypass MCMC for spike-and-slab priors (Ročková and George, 2014), we leverage the L_0 MM approximation as a fast, deterministic tool to bypass the combinatorial complexity of best subset selection.
- True MM for GLMs:** We replaced local PIRLS with a global quadratic majorizer for the logistic loss, ensuring true global majorization.
- Hierarchical Extension:**We extended the L_0 surrogate to grouped variables and Generalized Additive Models (GAMs), offering an alternative to spike-and-slab group lasso (Bai et al., 2022).

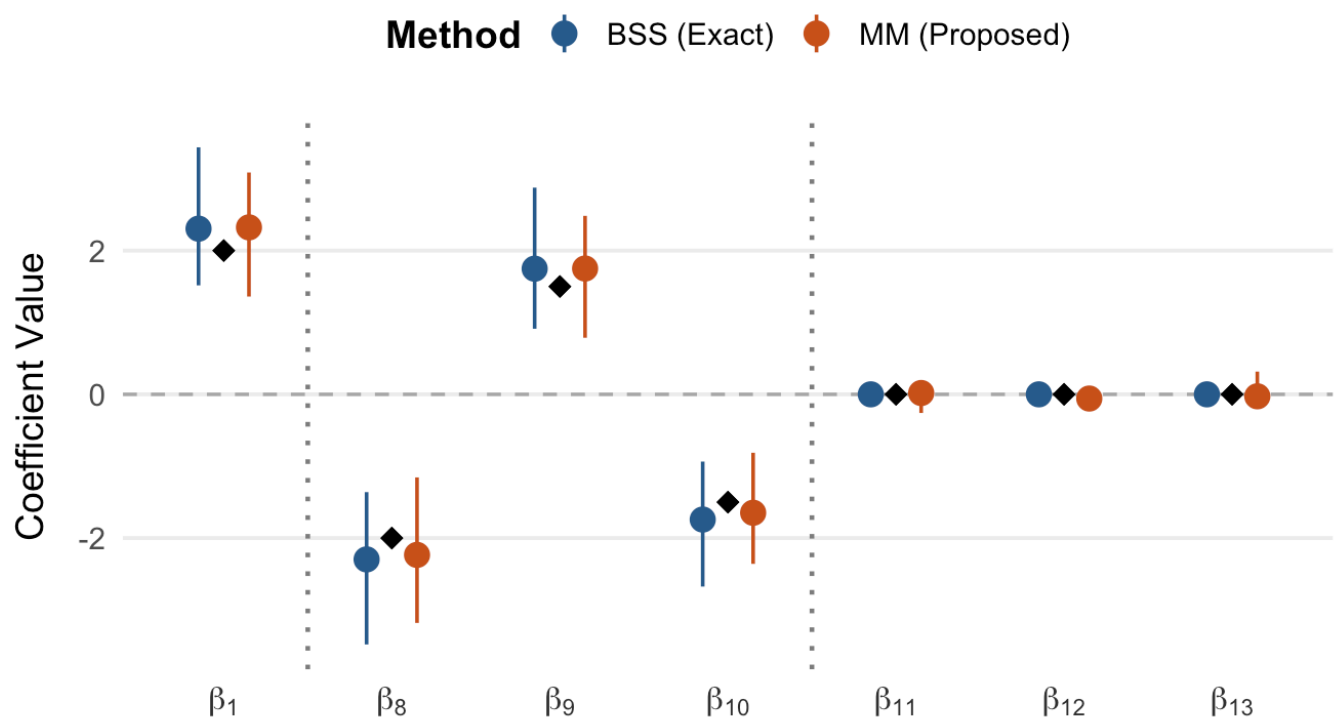


Figure 1. Binary simulation results. Comparison of BSS and MM coefficient estimates versus true values (black diamonds).

Statistical Setup

Let $y_1, \dots, y_n$ be independent observations from an exponential dispersion family

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}$$

where θ_i is the canonical parameter and ϕ is the dispersion parameter. The mean $\mu_i = b'(\theta_i)$ is connected to the linear predictor $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ via the link function $g(\mu_i) = \eta_i$. We estimate $\boldsymbol{\beta} \in \mathbb{R}^p$ by solving

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \{-\ell(\boldsymbol{\beta}) + \lambda q(\boldsymbol{\beta})\}$$

where $\ell(\boldsymbol{\beta})$ is the log-likelihood, $q(\boldsymbol{\beta})$ is the penalty function, and λ is the regularization parameter.

Structured L_0 Penalties

- Group L_0 :** Partition the predictors into G disjoint sets $\{S_g\}_{g=1}^G$. Analogous to the Group Lasso (Yuan and Lin, 2006), the penalty is:

$$q(\boldsymbol{\beta}) = \sum_{g=1}^G \mathbb{I}(\|\boldsymbol{\beta}_{S_g}\|_2 \neq 0)$$

- Functional L_0 :** For Generalized Additive Models with continuous predictors expanded via basis functions $f(x) = \boldsymbol{\psi}(x)^\top \boldsymbol{\gamma}$, the inclusion of the entire non-linear effect is penalized as a single group:

$$q(f) = \mathbb{I} \left(\sup_x |f(x)| \neq 0 \right) = \mathbb{I}(\|\boldsymbol{\gamma}\|_2 \neq 0)$$

Majorization-Minimization (MM) Framework

1. Majorizing the Penalty: From Lasso to Group Lasso

At each iteration k , given the current estimate $\boldsymbol{\beta}^{(k)}$, the non-convex penalty is upper-bounded by a convex surrogate function $Q(\boldsymbol{\beta} \mid \boldsymbol{\beta}^{(k)})$.

	Surrogate $Q(\boldsymbol{\beta} \mid \boldsymbol{\beta}^{(k)})$	Weights $\delta^{(k)}$	Estimation
Yen (2011)	$\lambda \sum_{j=1}^p \delta_j^{(k)} \beta_j + c$	$\frac{1}{\log(1 + \tau^{-1})(\beta_j^{(k)} + \tau)}$	Iterative Adaptive Lasso (Coordinate Descent)
Group Extension	$\lambda \sum_{g=1}^G \delta_g^{(k)} \ \boldsymbol{\beta}_{S_g}\ _2 + c$	$\frac{1}{\log(1 + \tau^{-1})(\ \boldsymbol{\beta}_{S_g}^{(k)}\ _2 + \tau)}$	Iterative Adaptive Group Lasso (Blockwise Coord. Descent)

2. Majorizing the Likelihood: A True MM for Logistic GLMs

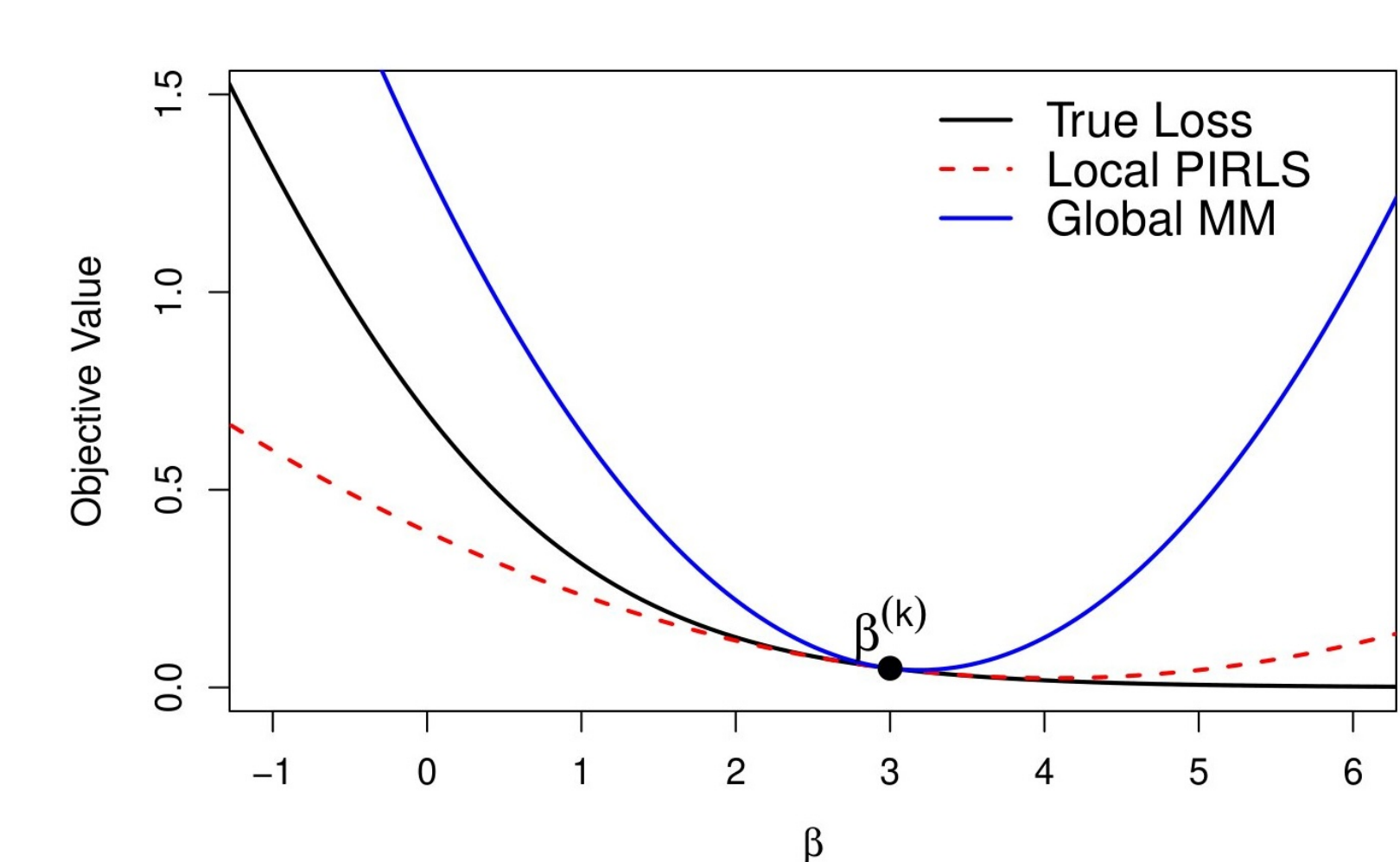


Figure 2. Surrogate functions for logistic loss.

- Yen's GLM Approach:** Yen (2011) employed PIRLS, which relies on the local Hessian. This provides only a local approximation, failing to guarantee global majorization.
- Our True MM Approach:** The logistic log-likelihood possesses a uniformly bounded curvature ($b''(\theta) \leq 1/4$). By leveraging the quadratic upper bound principle, we replace the local Hessian with this global bound (Figure 2).

Role and Calibration of the Regularization Parameter λ

Unlike standard penalties, L_0 **does not induce shrinkage**. Therefore, λ acts strictly as a sparsity threshold on the log-likelihood scale: a group is retained if and only if its inclusion improves the model fit by at least λ .

Optimal Tuning Strategies:

- Gaussian Case:** Tuned as a fraction of the residual deviance.
- Non-Gaussian Case:** Calibrated via a BIC-like threshold ($\lambda = \log(n)/2$).
- Data-Driven:** Cross-validation (CV) ensures robust out-of-sample generalization.

Numerical Study

1. AIM

Assess discrepancy between the proposed **MM** and the exact **BSS** optimum.

2. DESIGN

$\Rightarrow n = 200, p = 22$ (12 groups).
Correlated X ($\rho = 0.5$).
Gaussian and binary (logit) responses.
1000 replications, $\lambda = \log(n)/2$.

3. RESULTS

$\Rightarrow$ **MM** perfectly matches the **BSS** global optimum (see Figures 1 and 3).

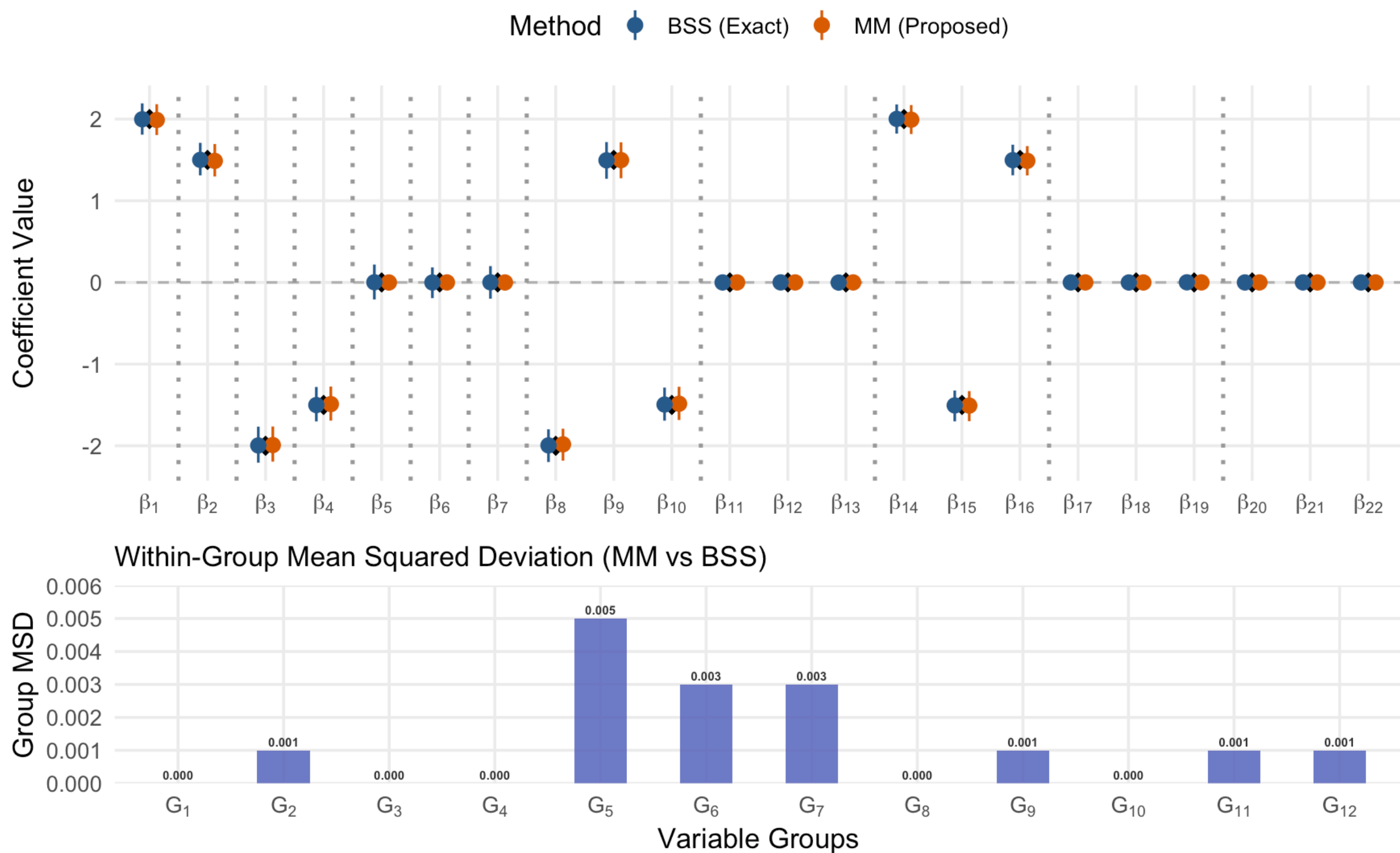


Figure 3. Gaussian simulation results. Top panel: BSS and MM coefficient estimates versus true values (black diamonds). Bottom panel: within-group mean squared deviation between BSS and MM.

Real Data Applications

German Credit Risk

Description: $n = 1000$ observations, $p = 20$ covariates (13 categorical, 7 numerical expanded via spline bases). Binary response indicating good/bad credit risk.

Aim: Compare group selection and the estimated functional effect of a continuous predictor between Group L_0 (MM) and Group Lasso (L_1).

Student Performance

Description: $n = 395$ students, $p = 30$ covariates (grouped into 31 blocks, with numerical variables expanded via splines). Response: final grade (treated via Gaussian regression).

Aim: Assess whether Group L_0 achieves comparable predictive accuracy to Lasso while retaining fewer, more interpretable groups.

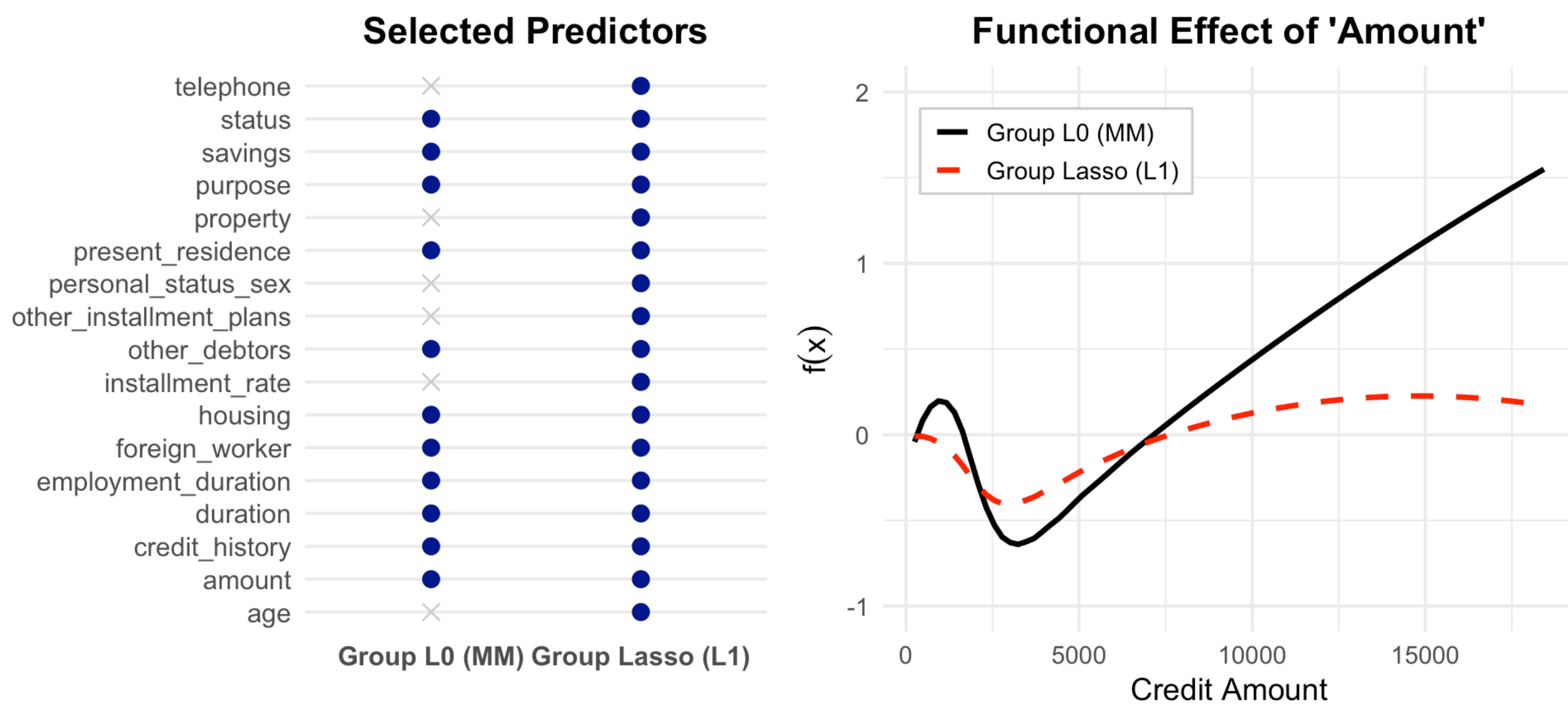


Figure 4. Selected predictors and estimated functional effect of *Amount* for Group L_0 (MM) vs. Group Lasso (L_1).

Group	$\ \boldsymbol{\beta}_{L_0}\ _2$	$\ \boldsymbol{\beta}_{L_1}\ _2$	Group	$\ \boldsymbol{\beta}_{L_0}\ _2$	$\ \boldsymbol{\beta}_{L_1}\ _2$	Group	$\ \boldsymbol{\beta}_{L_0}\ _2$	$\ \boldsymbol{\beta}_{L_1}\ _2$
failures	5.80	2.90	Medu	3.43	0.68	absences_ind	2.46	2.32
goout	0.00	0.76	age	0.00	0.75	freetime	0.00	0.67
higher	0.00	0.61	schoolsup	0.00	0.61	Mjob	0.00	0.54
sex	0.00	0.35	studytime	0.00	0.34	famsup	0.00	0.30
romantic	0.00	0.27	address	0.00	0.24	reason	0.00	0.24
famsize	0.00	0.21	internet	0.00	0.18	Dalc	0.00	0.15
Walc	0.00	0.13	Fjob	0.00	0.08	traveltime	0.00	0.07
Pstatus	0.00	0.04	health	0.00	0.03	activities	0.00	0.02
CV MSE	17.22	17.14	+ 7 additional groups with norm < 0.01					

Table 1. Group-wise coefficient norms for Group L_0 (MM) vs. Group Lasso (L_1) on Student Performance.

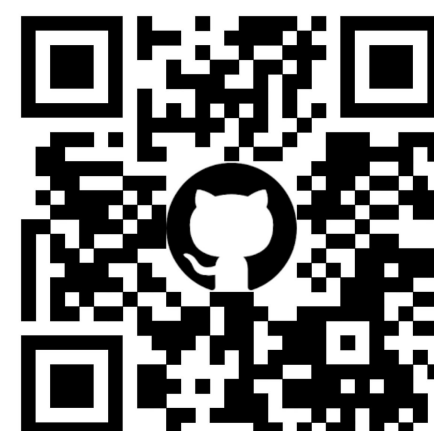
Conclusion: As shown in Figure 4, Group L_0 removes 5 uninformative groups that Lasso retains. It also accurately recovers the non-linear effect of *Amount* without the shrinkage bias of L_1 penalties.

Conclusion: As shown in Table 1, Group L_0 selects only 3 out of 31 groups (*Medu*, *absences_ind*, *failures*). It matches the CV MSE of Lasso while simplifying the model.

Take-home message

“Standard penalties like Lasso merge variable selection and shrinkage, forcing a trade-off between bias and false positives.”

Our Solution: By leveraging a globally bounding Majorization-Minimization algorithm, **Group L_0 regularization** achieves exact sparsity and unbiased estimation for GLMs, selecting features **without shrinking the signals**.



Scan for code

References

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